Measuring temperature from afar: Using Infrared Readings and Machine Learning Methods to Predict Oral Temperature

Introduction:

Quick and accurate disease identification is crucial in mounting an appropriate and effective response. Early identification is associated with a better response (Reyes et al., 2018), however, new outbreaks may not have adequate testing technologies, and therefore a proxy must be used instead. Elevated temperature is a common symptom of many diseases including COVID-19, where 98% of individuals within an observational study showed feverish symptoms (Yang et al., 2020). In healthcare the most common temperature reading methods in healthcare are oral and rectal, with the latter being the "gold standard" but suffers from its highly invasive nature (Medical Devices, 2024). However, both measurements involve physical contact and so contain a potential risk for cross contamination, an issue especially relevant when attempting to contain a new disease or strain. This has resulted in infrared measurements being widely used as an alternative. In this report I will outline the application of the machine learning methods to predict oral temperature based on various facial infrared measurements. My primary focuses will be on both its overall accuracy and how it handles older individuals, as they are more susceptible to disease.

To accomplish this, I will be using the dataset provided by Wang et al., (2020) from their similar study on the efficacy of infrared readings to measure elevated body temperatures. The data is comprised of infrared measurements taken from various locations around the individuals face (Figure 1). Four infrared readings were taken at every facial location for every individual, and the humidity, cosmetics, and distance. It also contains two types of oral temperature, differentiated by the mode that was used on the recording apparatus (fast/monitoring). In my modelling I will use aveOralM as the target variable, temperature recorded under monitor mode as this gives more accurate measurements. For modelling I will employ both a linear regression and neural network model, due to their conflicting benefits in capturing relationships within the data. Regression models are suited to capturing linear relationships between the predictor and the target variables, whereas neural nets can capture non-linearities in the data. To do this I will employ and test two different types of neural networks: a multi-layer perceptron (MLP), and recurrent neural network (RNN). The former will be able to capture complex relationships, and the latter will add to this by capturing temporal relationships observed between the readings at separate intervals.

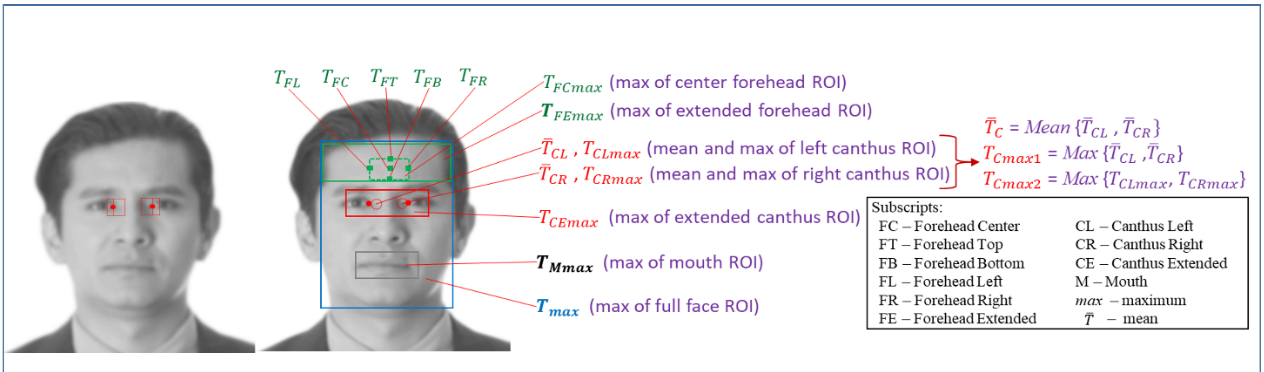

Figure 1. Delineated facial regions and critical points on thermal images: forehead regions and points (green), canthi region and points (red), mouth region (gray rectangle), and entire face (blue rectangle).
Note: The above image is a generic face (based on PowerPoint clip art: Insert > Icons > Cutout People >Alfredo) used for illustration purposes and not an actual participant in our study.

Figure 1. Location of the infrared measurements and their name. (Source: Zhou et al., (2020), Wang et al., (2021), Ghassemi et al., (2018), Chenna et al., (2018))

Data Cleaning

The provided data is comprised of two different datasets, FLIR and ICI which differ in the type of infrared camera used. To determine if one set of data was of higher quality than the other, I transformed the time series data into feature mean data, omitting all missing values. Both ordinary least squares regression models were practically identical (Table S1), implying similar quality. Though when observing the missing data counts for each of the datasets, ICI contained far more NA values than FLIR (Figure 2). For this reason, I decided to omit the ICI dataset and rather exclusively use FLIR.

The data contains an approximate missingness rate of 1.5% (Figure 3), and the trends of which suggest that the type of missingness was not Missing Completely at Random (MCAR) (Figure 2). To investigate whether missingness was associated with the outcome variable, we conducted Wilcoxon rank-sum tests comparing the oral temperature values between the NA group, and non-NA group, for every predictor variable. The results indicated that, for a subset of variables, the outcome values differed significantly between the missing and non-missing groups ($p < 0.05$) (Table S2). This suggests that the data are not MCAR. Rather, the pattern of missingness is consistent with Missing at Random, or Missing Not at Random, depending on the specific variable and context. As a result, complete case analysis should be avoided due to the bias introduced by excluding s specific subset of information, however, I was unable to perform feature imputation using MICE, due to the package being unavailable. Rather I have performed median value imputation to retain all other useful information.

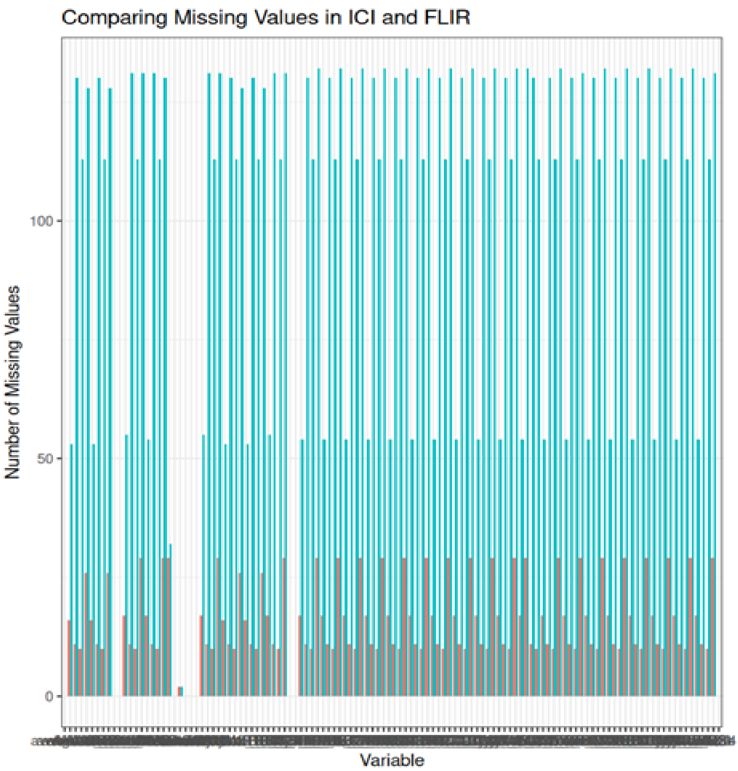

Figure 2. Distribution of missing values for FLIR and ICI datasets.

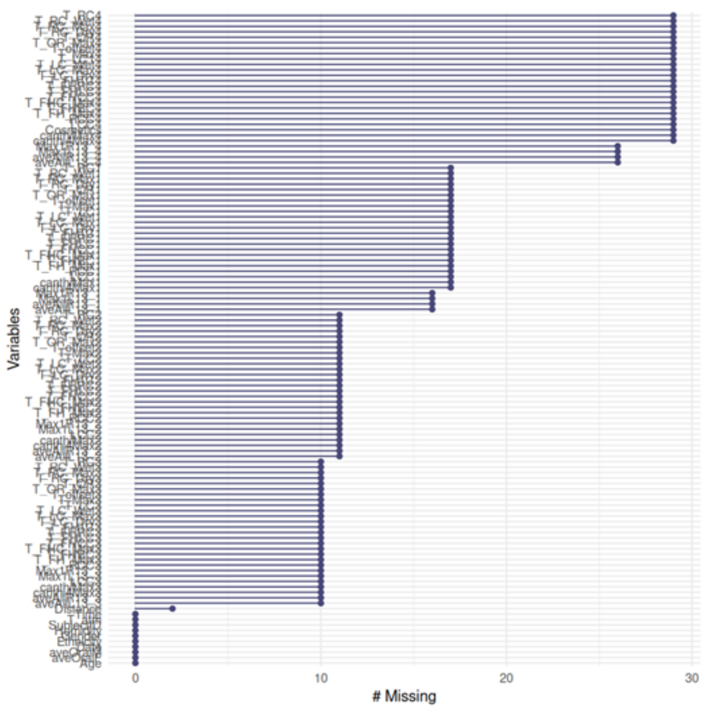

Figure 3. Missing values per variable in the FLIR data.

Exploratory Data Analysis

Infrared Measurement Data

The distributions of the infrared measurements can all be seen in Figure S1. All features are non-normally distributed, so I will apply non-parametric tests to my data analysis. These measurements showed many extreme values, though none showed any valid reasons for removal, due to incorrect data inputting etc. Many of these likely contain important information of individuals suffering with disease and fever and thus will be useful to my models. The temporal nature of my infrared data appears to have an influence on the infrared features. Some variables show trend over time (Figure 3), and all variables showed significant differences over time. This suggests that a model that can understand and use temporal information may be advantageous. Although there are no strong linear correlative relationships between time and feature therefore this may not be captured in a linear model.

Many mean infrared measurements show high levels of multicollinearity (Figure 4), which is expected given the proximity of them. Further, the presence of features for both average and maximum temperatures within regions, likely introduces a lot of redundancy. The dependent variable is also well correlated with the infrared measurements, with the whole face, oral/mouth, and canthus region being the most strongly correlated (Figure 5). This supports the selection of a regression model as they excel in handline linear relationships.

Similarly, the non-infrared continuous data is too non-normally distributed, including oral temperature (Figure S3). Analysis of these highlights potential sampling bias within our data. The distributions of Age, Ethnicity, and Gender all have important implications when considering the use of these models as a proxy to measure disease. Specifically, younger individuals are overrepresented while older age groups are comparatively sparse, with gender and ethnic group also being skewed (Figure S2). In a clinical or public health setting, this lack of demographic diversity can bias a model's predictions, leading to reduced accuracy for underrepresented populations. To address this when testing the models we will explicitly test against these minority groups to ensure equal predictive ability.

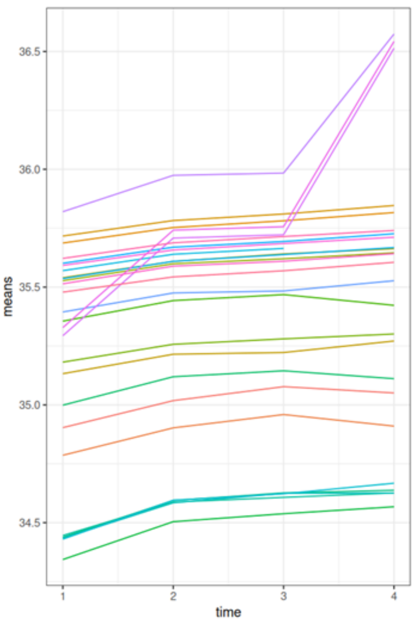

Figure 3. Temporal changes of all mean infrared variables.

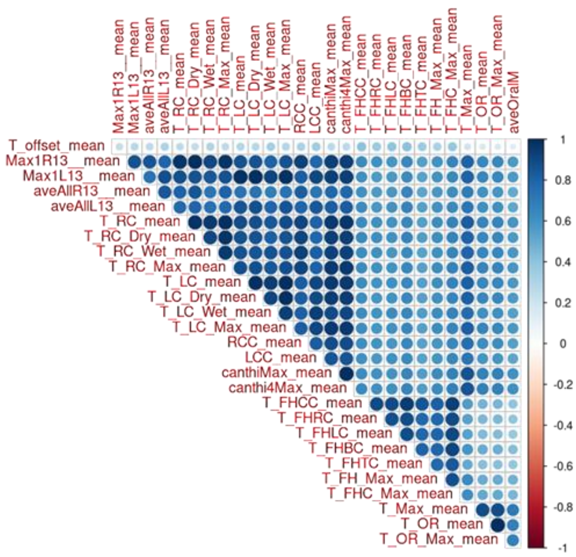

Figure 4. Correlation matrix of all mean infrared variables. Correlation scores calculated using Spearman's rank.

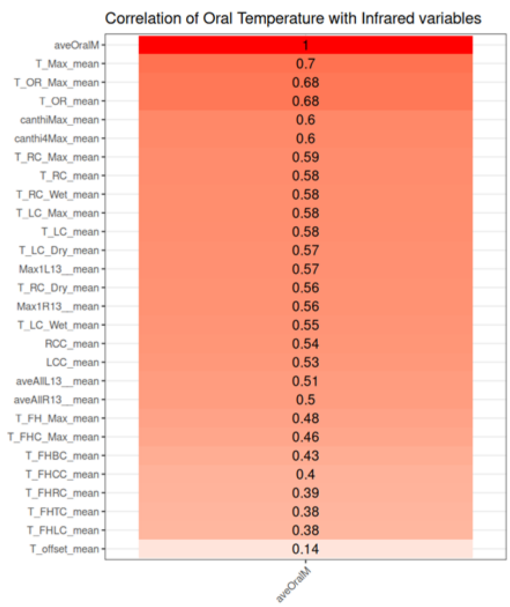

Figure 5. Correlation scores of mean infrared measurements and oral temperature. Calculated with Spearman's Rank.

Feature Permutation

My data contains many features (24), and many likely hold little predictive power for oral temperature. Consequently, I performed feature permutation to identify the most important features prior to data processing. It involves shuffling each variable in the test dataset, and observing what variables cause the largest increase in mean squared error (MSE). The primary benefit of which, is it being model agnostic, and its application to different models allows for a unique selection of features that fit the benefits of that specific model. Regression models are beneficial when modelling linear relationships, resulting in the variables in Figure 6 being deemed “important.” Whilst neural networks (NN) capture complex relationships due to non-linear activation functions between layers. I used a random forest model to determine which features are important. In random forests, ensembles of decision trees naturally identify splits that interactively segment the feature space, while neural networks learn layered representations that capture non-linearities in a similarly flexible way (Figure 7).

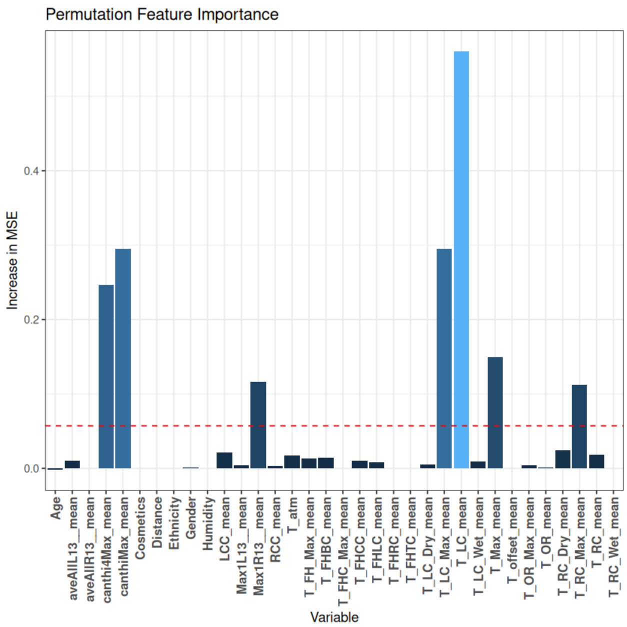

Figure 6. Increase in MSE per variable during regression permutation. The red line represents the mean increase in MSE.

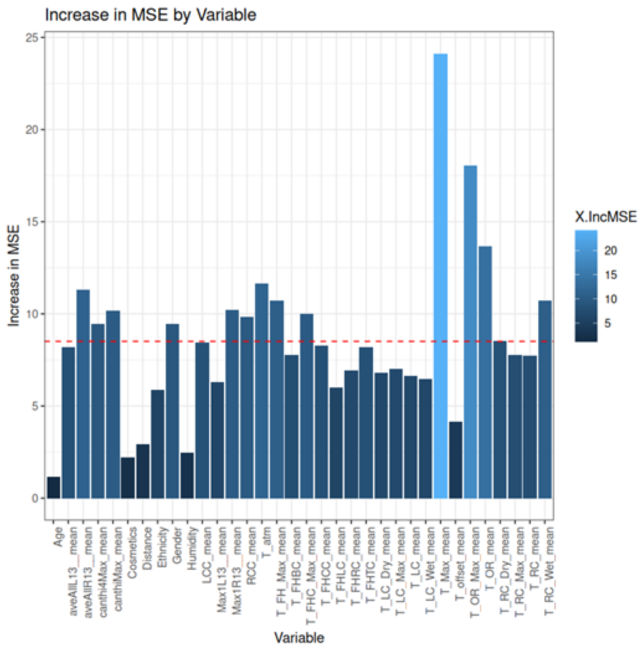

Figure 7. Increase in MSE per variable during random forest permutation.

Principal Component Analysis

These selected feature sets exhibited substantial multicollinearity and redundancy (Figure 8), which can inflate variances and degrade interpretability in regression models and similarly lead to unstable training or overfitting in neural networks. To address this, I applied Principal Component Analysis (PCA), a linear transformation that projects the original correlated features into a smaller set of uncorrelated principal components (PCs). I retained only the components that captured >95% of the variation in the data (Linear regression = 2, NN = 4, Figure X). In regression, this mitigates the issue of multicollinearity (Figure 9) resulting in more reliable coefficient estimates. For neural networks (NN), PCA simplifies the feature space so that training is more stable and less prone to overfitting on redundant signals. The data was transformed so that the PCs incorporated temporal information ensuring maintenance of time-based dependencies, with this method having has been applied to time-series extrinsic regression NN models (Gao et al., 2024).

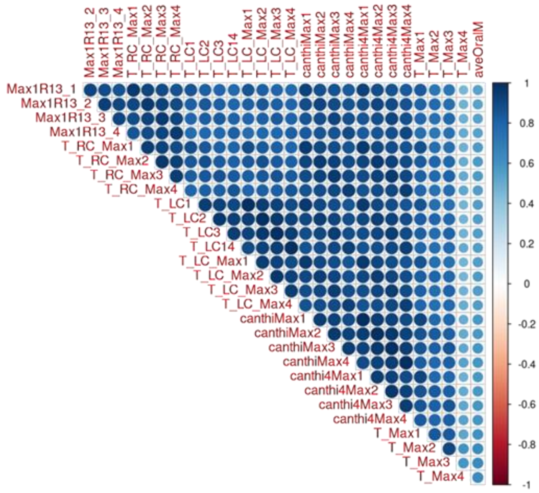

Figure 8. Correlation matrix of important features from regression permutation.

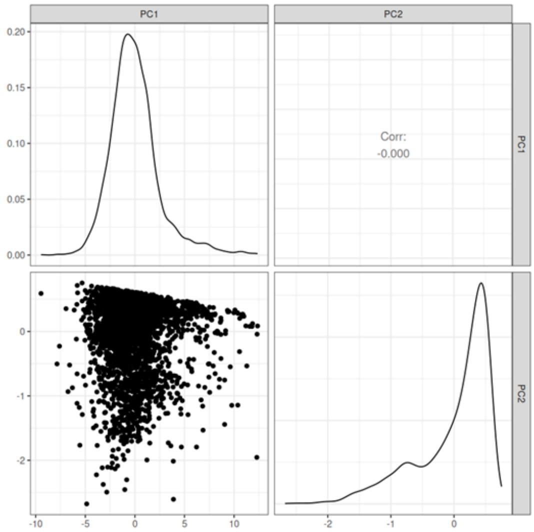

Figure 9. Scatterplot matrix of the two selected PCs for the regression model. The diagonal represents the kernel density estimation.

Model 1 – Regression

As previously mentioned, regression was selected due to the correlated nature of my dataset. However, OLS models are often not applied to the real world due to concerns with overfitting. This is the process in which the model fits too closely to the training data, and its noise, rather than the underlying trends. This results in a model that performs well in theory but poorly when applied in the real world. This encompasses the need for regularisation, the introduction of a penalty to the loss function, to mitigate complex models and large coefficients. In regression this is done via L1 and L2 regularisation.

L2 (ridge) regularisation, involves the application of a constant, λ, that is multiplied by the sum of the coefficients *(*β) squared. This is beneficial for generalisation by mitigating the impact of noisy features, through the shrinkage of their features. L1 (Lasso) regularisation, works similarly. The primary difference is λ being multiplied by the sum of the absolute values of β. This penalisation parameter introduces a constraint that the values of the coefficients can take, with the shape varying depending on the type. Ridge (B^2) penalisation creates a spherical constraint, whereas Lasso (|β|), imposes a diamond-shaped constraint. These constraints allow for Lasso to zero out coefficients while Ridge shrinks them uniformly. For both models as λ increases as does the strength of the penalty, causing more shrinkage to the coefficients.

I have elected to utilise an elastic net model as this method will find the optimal balance of the two methods. This method utilises a loss function that is a combination of L1 and L2. This is controlled by the coefficient, α, otherwise known as the L1 ratio.

Hyperparameter Selection

To calculate the optimal values for λ, and α, I employed K-fold cross validation. This is a sampling technique which divides up the set of observations in the training data in K equal sized folds. You then iteratively assign a single fold as the test set and train the model using the remaining K-1 sets. Following this you average out the loss between all hyperparameters you tested and find the optimal values. I performed cross validation with K = 10, with the data randomly shuffled prior to fold creation. This provides a reliable estimate of model performance by using all available data. Through this process, I identified α=1.0 and λ=0. 0025 as optimal, achieving an RMSE of 0.269 (Figure 9). This is equivalent to having performed Lasso and has resulted in the removal of PC1 at the third timestep (Table S3). By combining PCA for dimensionality reduction with elastic net regularisation, the final model is less prone to overfitting, and better able to generalise to real-world data.

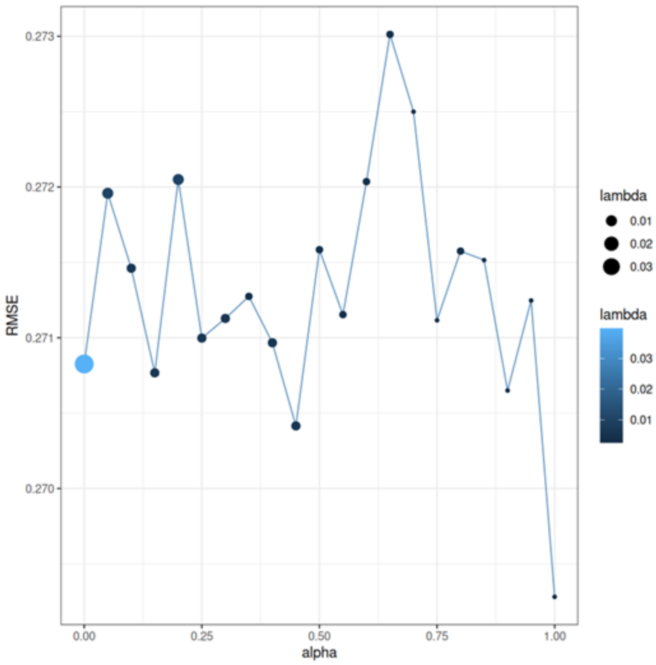

Figure 9. Scatterplot matrix of the two selected PCs for the regression model. The diagonal represents the kernel density estimation.

Neural Network

Multilayer Perceptron

A Multilayer Perceptron (MLP) is a type of feedforward NN, in which all neurons are fully connected with non-linear activation functions between neurons. The training process involves prediction using initially random parameters, and the subsequent calculation of error, in my case I am using mean-squared error (MSE). This is then followed by a process termed back-propagation. This involves propagating the error backwards through the NN, and by using the chain rule, returns the gradient for how weights and bias should be adjusted to decrease the loss. The way the optimisation function does this depends on the equation you use. In this instance I used the Adam optimiser, a stochastic gradient method that computes adaptive learning rates for different parameters using the first and second order estimates of the gradients (Kingma & Ba, 2014). This mitigates the problems from many alternate optimisation algorithms. The addition of a non-linear ReLU activation function, both introduces non-linearity, and mitigates the vanishing gradient problem. My MLP contained 1 hidden layer, with 15 nodes, as to limit the number of internal parameters (316), as my training data was relatively small (n=652).

Recurrent Neural Networks

Recurrent neural networks (RNNs) are an alternative type of NN that can incorporate temporal information, via the use of neural "loops" (Figure X). This circular nature allows for information from the previous timestep to be passed back into the network (Idrees, 2024), this "memory" may be useful due to the time dependencies in the data (Figure X). In trains in a similar manner to MLP, except the occurrence of backpropagation through time, which involves RNN unfolding propagating errors across time steps to adjust the network (GeeksforGeeks, 2018). This process, however, is computationally expensive due to the computation of gradients at each time step, causing memory limitations and slow training. I chose to employ a Gated Recurrent Unit, due to their relative simplicity and speed when compared to Long Short-Term Memory Networks, despite worse performance (Idrees, 2024). The time dependent nature of this model meant I had to exclude the non-temporal information.

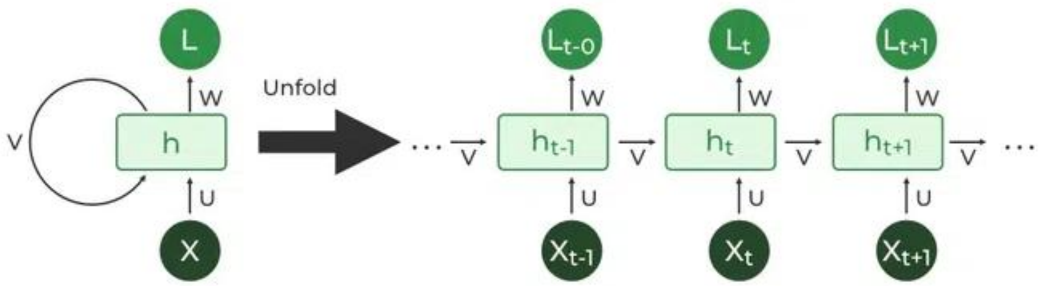

Figure 10. Basic framework of a recurrent neural network. (Source: GeeksforGeeks, 2018)

Hyperparameter Selection

Rather than using the computationally expensive K-fold cross validation for hyperparameter selection I used simple error calculation upon a test dataset (n=163). Learning rate represents the step size of the coefficients within optimisation. If too large you can "jump" either side of the optimal parameter values whereas if too small, it can converge too slowly. I selected 0.067 for my MLP, as though it did not converge the quickest, I do not want to miss the optima, for RNN I selected 0.045 (Figure S4). Batch size is the number of rows of data processed by the model before the parameters are updated. This is a trade-off between convergence and cost. Large sizes are also prone to overfitting as there is less pronounced noise in every batch, due to the law of large numbers, with this allowing it to more easily fit perfectly to the training data. The difference in convergence speed can be seen in Figure 11/12. I chose values of 32 for both models, due to quick convergence and stability. Dropout represents the primary form of regularisation for NN which involves randomly ignoring a proportion of the neurons by setting their coefficients to 0. This forces the model to become more robust and generalisable rather than learning the specifics of the dataset, and the optimal value represents a trade-off between over and underfitting. MLP loss = 0.1, RNN = 0.1(Figure S5). The training of both these models did not suggest that overfitting occurred (Figure S6).

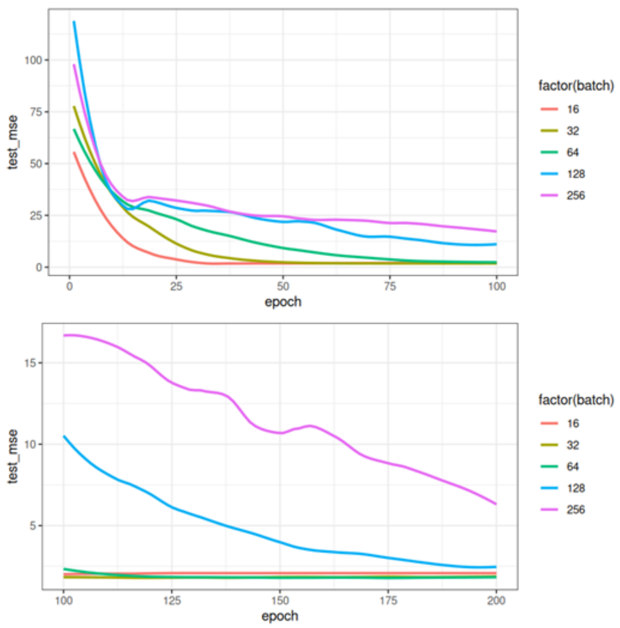

Figure 11. Test error curves of MLP for different values of batch size.

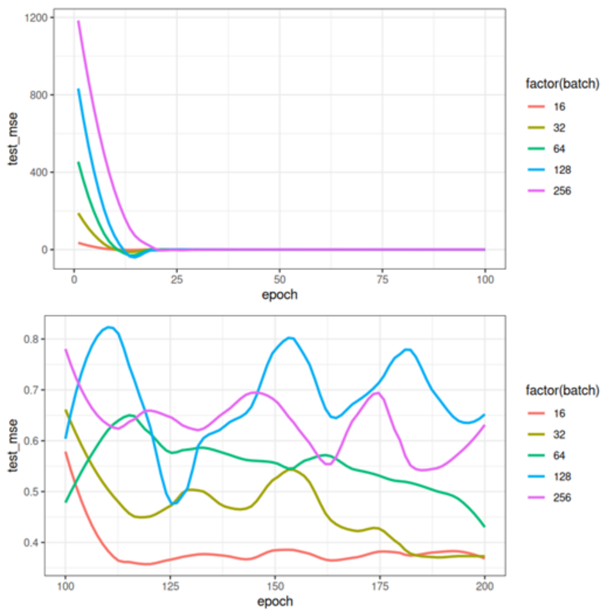

Figure 12. Test error curves of RNN for different values of batch size.

<u>Model Comparison</u>

Model performance was evaluated using a dedicated validation set (n=204) to ensure robust and unbiased comparisons. Data were initially partitioned into training and validation subsets, thereby enabling each model to learn and refine its parameters using only the training portion. Once trained, these models were assessed on the reserved validation set, which had remained unseen during the training phase. This approach reduces the risk of overfitting and provides an independent measure of generalisability.

Overall, the elastic net and MLP models both performed well, with the MLP performing the best with a mean absolute error (MAE) of 0.058 (Table S4). This suggests that the NN was able to utilise non-linear interactions within the data to predict oral temperature well. Though the similar regression results, suggest that the linear features are equally important. To test this, I created a "linear MLP" model, where all parameters remained constant, except for the removal of the non-linear activation function. The similarity of this to the MLP suggests that it may have been the NN architecture that resulted in improved performance, rather than the non-linear interactions, but this model should not be used as it still was outperformed by the regression. The comparatively high error scores for the RNN are likely a symptom of the small dataset. The natural complexity of an RNN (parameters=2941) and the limited size of my training set (n=652), likely resulted in poor capturing of trends and underfitting.

To compare performance between minority and non-minority group I split the validation data into a cumulative minority group and non-minority group. The MLP model showed almost identical performance, regardless of group, whereas regression showed a drastic difference. This may suggest that linear trends between infrared data and oral temperature may differ in different minority groups, thereby causing it to model poorly. Due to the lack of data I was unable to make any prominent inferences in specific minority groups.

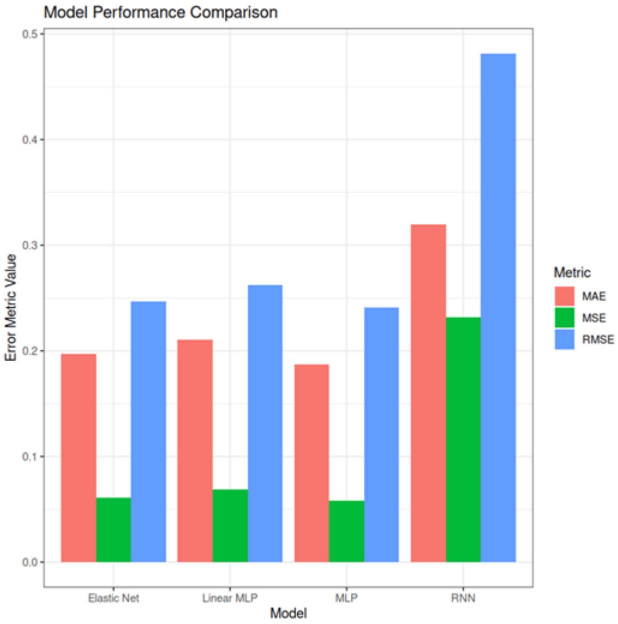

Figure 13. Error metric comparison for each of my models. MLP = Multi-Layer Perceptron, RNN = Recurrent Neural Network.

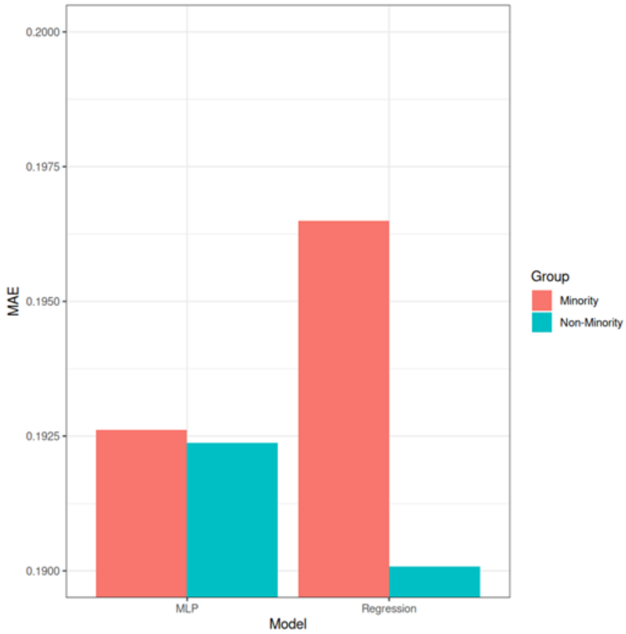

Figure 14. Performance comparison (MAE) of MLP and elastic net model when looking at minority vs non-minority groups within the validation data. (Minority included: non-white ethnicity, and age > 30).

Results and Conclusion

These comparative analyses suggest that, whilst a NN performs better, regularised linear approaches still tend to perform as well, at performing oral temperature using infrared readings. Though, this conclusion is aided by the small size of my initial dataset (n=1020), thereby limiting the quality of inferences made, and complexity of the MLP. Therefore, going forward research should focus on improving data quality and architecture for NN's. Modifications to the number of hidden layers, and nodes should be investigated, alongside the use of alternate activation functions (for example TanH, or leaky ReLU. Ideally, however, I believe resources should be focussed on increasing data collection, providing more data to improve our model on, with focus on both the quantity and quality of the data. Greater measurement of current minority groups should be taken, especially if this is to be employed in a medical setting. This should occur alongside the inclusion of further, potentially implicative measures, such as liquid ingestion.

Clinically, the reliable prediction of oral temperature can assist healthcare staff in identifying early warning signs of infection or fever onset, thereby informing timely interventions and more targeted patient monitoring. However, evidence has suggested that oral temperature is an unsuitable diagnostic tool for determining internal body temperature (Mazerolle et al., 2011), therefore a model predicting this temperature may prove redundant. Rather a model should be produced to predict rectal temperature without invasive measurements. Although, due to the nature of this data collection, the resulting dataset will likely be small, and therefore a linear regression model should be utilised due to its strong performance and ease of tuning on small, but linearly correlated datasets.

Supplementary Materials:

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)     4.3245098  0.6708165   6.447 1.78e-10 ***
T_offset_mean  -0.0558855  0.0258716  -2.160  0.03100 *  
Max1R13__mean  -0.3687350  0.1868869  -1.973  0.04877 *  
Max1L13__mean  -0.1703548  0.1578743  -1.079  0.28083    
aveAllR13__mean -0.0152544  0.0358264  -0.426  0.67036    
aveAllL13__mean -0.0624601  0.0440286  -1.419  0.15632    
T_RC_mean      -0.5557610  0.9466221  -0.587  0.55727    
T_RC_Dry_mean   0.2314353  0.1960630   1.180  0.23812    
T_RC_Wet_mean   0.0205904  0.1031942   0.200  0.84189    
T_RC_Max_mean   0.7906173  0.9011904   0.877  0.38053    
T_LC_mean       0.9683060  0.9066724   1.068  0.28579    
T_LC_Dry_mean  -0.1666461  0.2493290  -0.668  0.50405    
T_LC_Wet_mean  -0.1509835  0.0857365  -1.761  0.07854 .  
T_LC_Max_mean  -0.4879210  0.8382571  -0.582  0.56065    
RCC_mean        0.0500901  0.0675986   0.741  0.45887    
LCC_mean        0.1415969  0.0631172   2.243  0.02509 *  
canthiMax_mean -0.6592941  0.9878884  -0.667  0.50469    
canthi4Max_mean 0.5321090  1.0124416   0.526  0.59930    
T_FHCC_mean    -0.1275435  0.0524959  -2.430  0.01529 *  
T_FHRC_mean    -0.0009877  0.0361617  -0.027  0.97822    
T_FHLC_mean    -0.0943875  0.0320498  -2.945  0.00330 ** 
T_FHBC_mean     0.0825095  0.0492076   1.677  0.09390 .  
T_FHTC_mean     0.0204198  0.0236228   0.864  0.38757    
T_FH_Max_mean   0.1370622  0.0415737   3.297  0.00101 ** 
T_FHC_Max_mean  0.0540906  0.0593432   0.911  0.36226    
T_Max_mean      0.6583793  0.0755484   8.715  < 2e-16 ***
T_OR_mean       0.2357489  0.6948032   0.339  0.73445    
T_OR_Max_mean  -0.1533401  0.6931028  -0.221  0.82495    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2638 on 992 degrees of freedom
Multiple R-squared:  0.7391,	Adjusted R-squared:  0.732 
F-statistic: 104.1 on 27 and 992 DF,  p-value: < 2.2e-16
```

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)     4.3245098  0.6708165   6.447 1.78e-10 ***
T_offset_mean  -0.0558855  0.0258716  -2.160  0.03100 *  
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aveAllR13__mean -0.0152544  0.0358264  -0.426  0.67036    
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T_RC_mean      -0.5557610  0.9466221  -0.587  0.55727    
T_RC_Dry_mean   0.2314353  0.1960630   1.180  0.23812    
T_RC_Wet_mean   0.0205904  0.1031942   0.200  0.84189    
T_RC_Max_mean   0.7906173  0.9011904   0.877  0.38053    
T_LC_mean       0.9683060  0.9066724   1.068  0.28579    
T_LC_Dry_mean  -0.1666461  0.2493290  -0.668  0.50405    
T_LC_Wet_mean  -0.1509835  0.0857365  -1.761  0.07854 .  
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RCC_mean        0.0500901  0.0675986   0.741  0.45887    
LCC_mean        0.1415969  0.0631172   2.243  0.02509 *  
canthiMax_mean -0.6592941  0.9878884  -0.667  0.50469    
canthi4Max_mean 0.5321090  1.0124416   0.526  0.59930    
T_FHCC_mean    -0.1275435  0.0524959  -2.430  0.01529 *  
T_FHRC_mean    -0.0009877  0.0361617  -0.027  0.97822    
T_FHLC_mean    -0.0943875  0.0320498  -2.945  0.00330 ** 
T_FHBC_mean     0.0825095  0.0492076   1.677  0.09390 .  
T_FHTC_mean     0.0204198  0.0236228   0.864  0.38757    
T_FH_Max_mean   0.1370622  0.0415737   3.297  0.00101 ** 
T_FHC_Max_mean  0.0540906  0.0593432   0.911  0.36226    
T_Max_mean      0.6583793  0.0755484   8.715  < 2e-16 ***
T_OR_mean       0.2357489  0.6948032   0.339  0.73445    
T_OR_Max_mean  -0.1533401  0.6931028  -0.221  0.82495    
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2638 on 992 degrees of freedom
Multiple R-squared:  0.7391,	Adjusted R-squared:  0.732 
F-statistic: 104.1 on 27 and 992 DF,  p-value: < 2.2e-16
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Table S1. Comparison of ICI (left) and FLIR (right) ordinary least squares regression summaries.

```
[1] "Column T_offset2 has significant changes in NA and non NA groups (p-value = 0.00542425223163316"
[1] "Column Max1R13_2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
[1] "Column Max1L13_2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
[1] "Column aveAllR13_2 has significant changes in NA and non NA groups (p-value = 0.0218685673662579"
[1] "Column aveAllL13_2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
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[1] "Column T_RC_Dry2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
[1] "Column T_RC_Wet2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
[1] "Column T_RC_Max2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
[1] "Column T_LC2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
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[1] "Column T_OR_Max2 has significant changes in NA and non NA groups (p-value = 0.00330019761094022"
```

Table S2. Output of Wilcox test, highlighting variables where there is a difference in oral temperature between NA and non-NA groups.

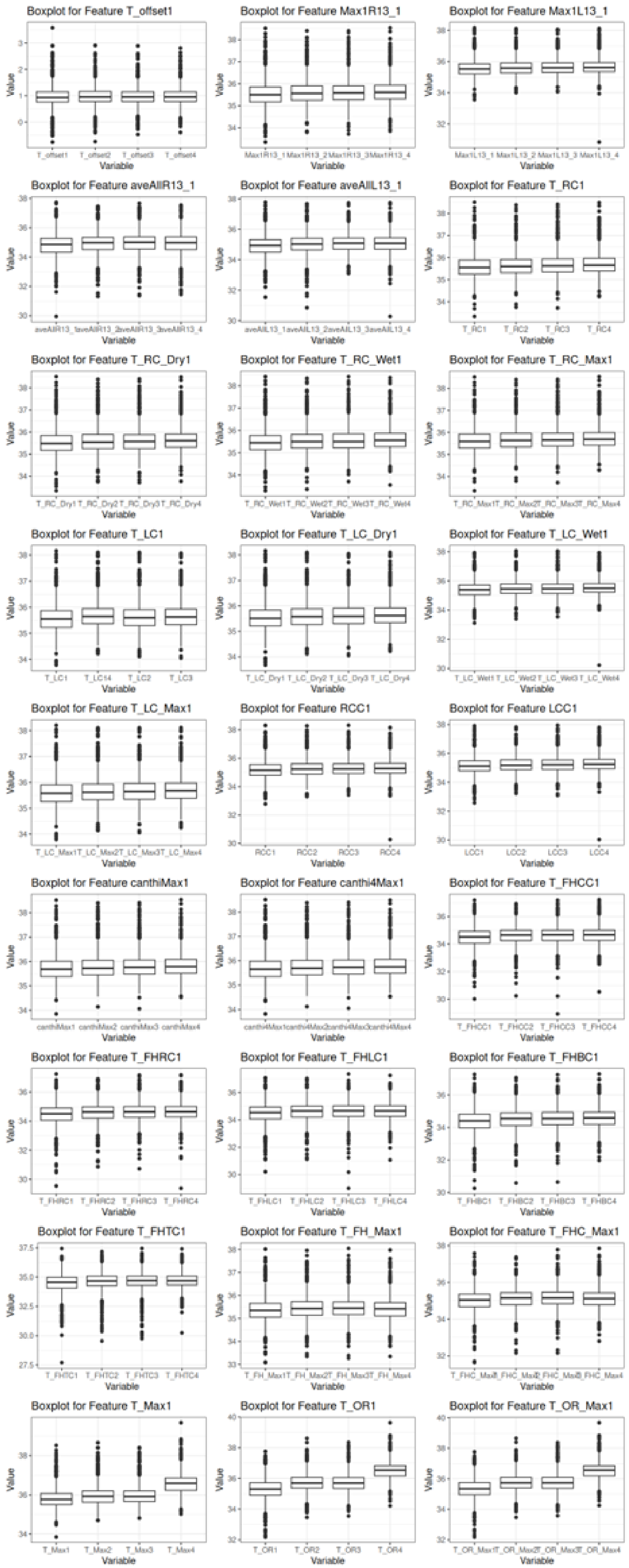

Figure S1. Boxplots of all infrared variables, split by the time.

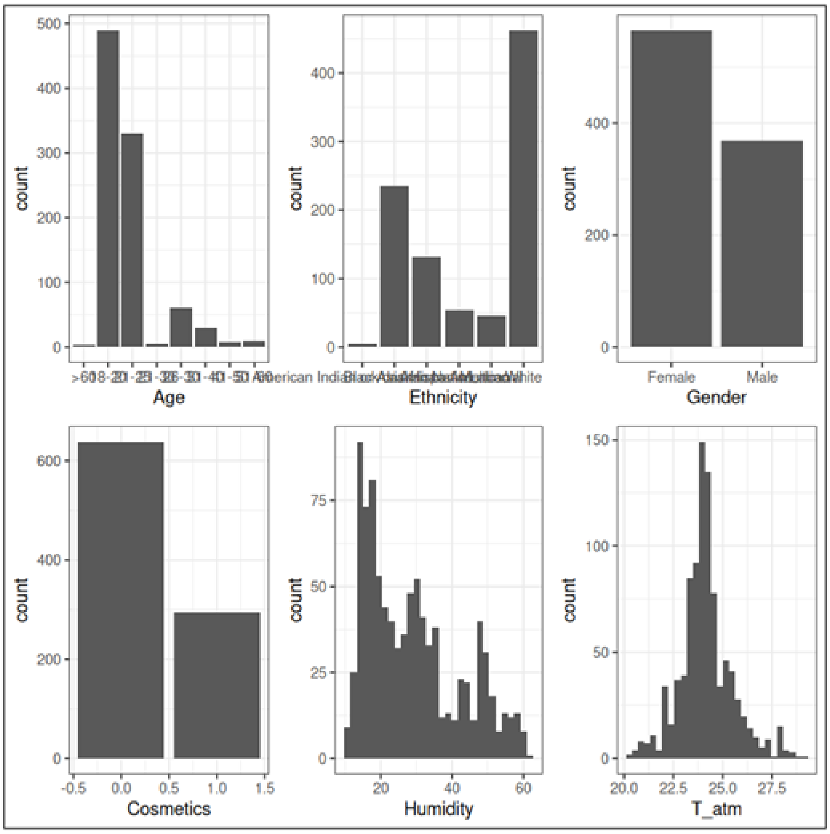

Figure S2. Distribution of all non-infrared data.

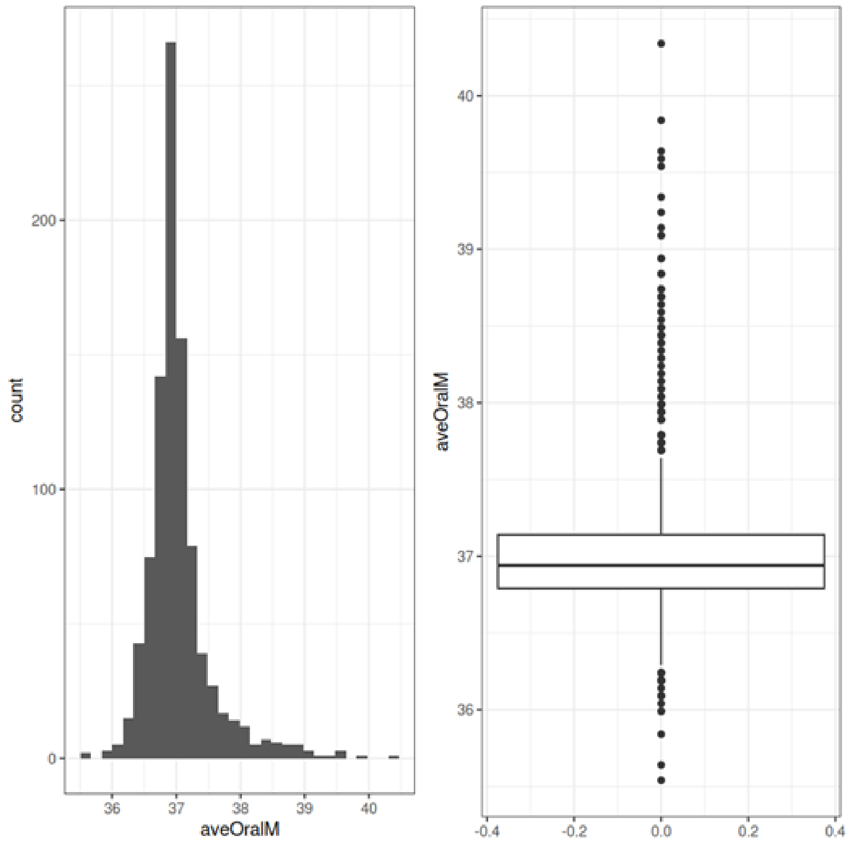

Figure S3. Distribution or Oral temp measured in monitor mode (aveOralM).

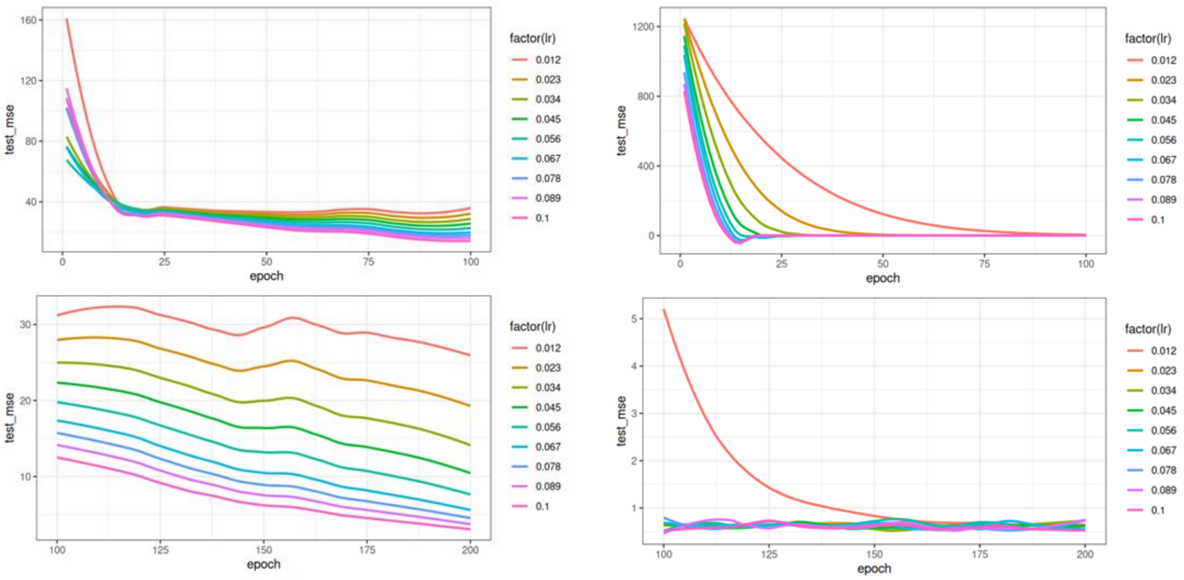

Figure S4. Test error curves for the training of MLP (left) and RNN (right) under different sizes for learning rate.

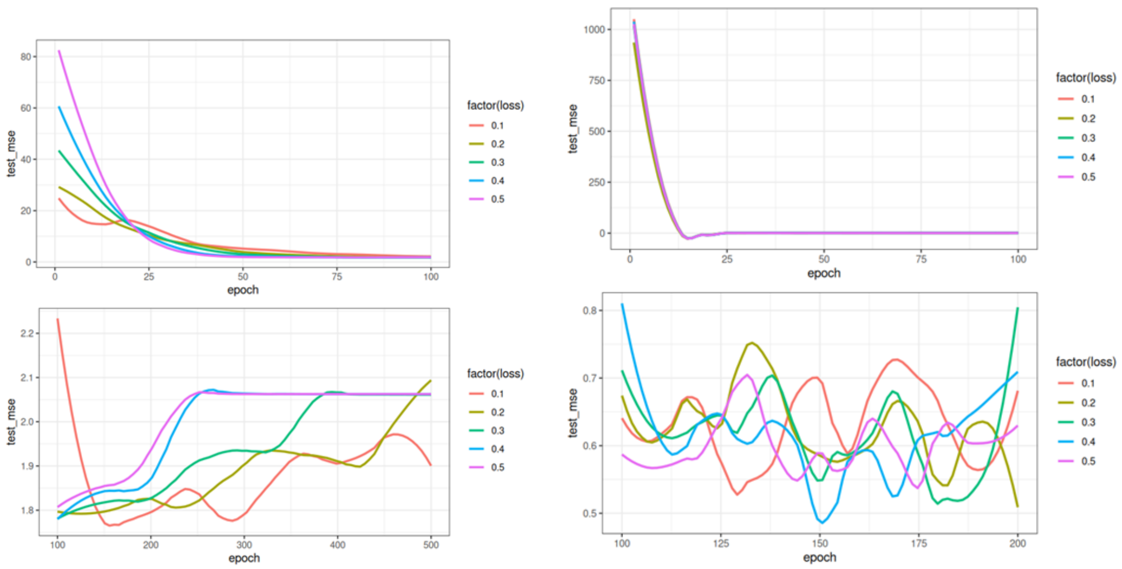

Figure S5. Test error curves for the training of MLP (left) and RNN (right) under different sizes for loss.

	s0
(Intercept)	36.95288691
PC1_1	0.04027818
PC2_1	-0.06796663
PC1_2	.
PC2_2	-0.07426478
PC1_3	0.07175854
PC2_3	-0.09817893
PC1_4	0.07266992
PC2_4	-0.17965596

Table S3. Coefficients derived from my elastic net regression.

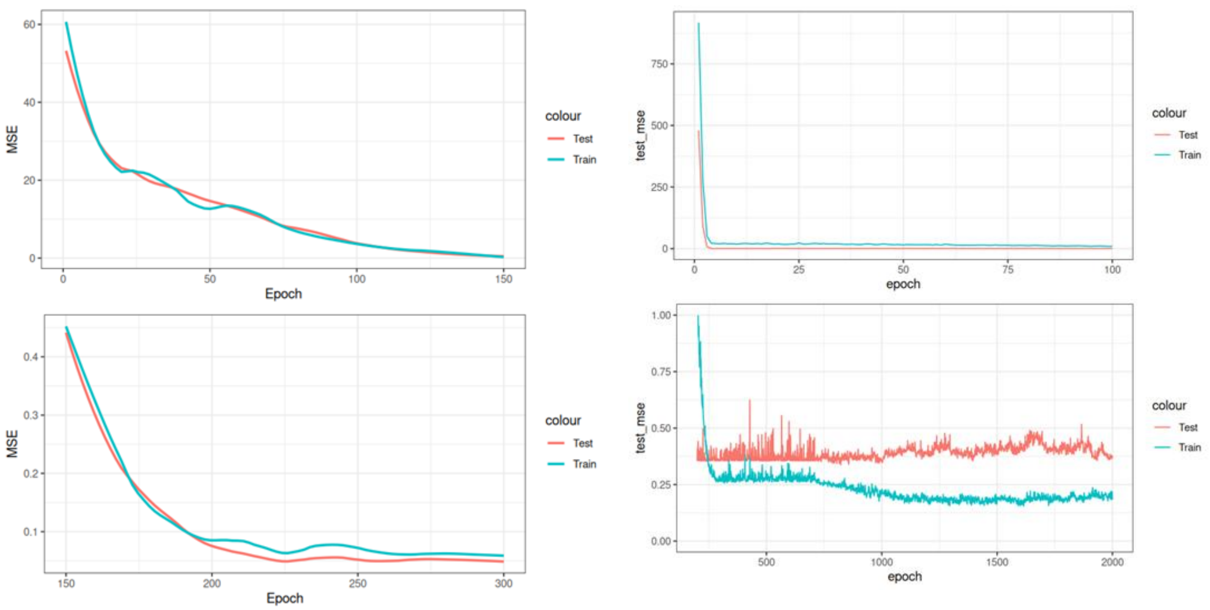

Figure S6. Error curves during training for my final MLP (left) and RNN (right) models. MSE was used and calculated on both my test and training datasets.

A tibble: 12 × 3

Model	Metric	Value
<chr>	<chr>	<dbl>
MLP	MSE	0.05806091
MLP	RMSE	0.24095832
MLP	MAE	0.18729673
RNN	MSE	0.23167688
RNN	RMSE	0.48132825
RNN	MAE	0.31973034
Linear MLP	MSE	0.06864917
Linear MLP	RMSE	0.26200987
Linear MLP	MAE	0.21029459
Elastic Net	MSE	0.06092065
Elastic Net	RMSE	0.24682109
Elastic Net	MAE	0.19677387

Table S4. Final error metrics for all my models.

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